

# Liam Wetzel

lsw59@case.edu | liamswetzel@gmail.com

## Education

### Case Western Reserve University

Doctorate of Philosophy, Systems Biology and Bioinformatics (current)

Masters of Science, Pathology (May 2023), Cumulative GPA: 3.79

### University of Nevada, Las Vegas

Bachelors of Biological Science, Minor Chemistry (May 2019)

Bachelors of Psychology, Minor Neuroscience (May 2019), Cumulative GPA: 3.41

## Publications

Liu, T., **Wetzel, L.**, Zhu, Z., Kumaraguru, P., Gorthi, V., Yan, Y., Bukhari, M. Z., Ermekbaeva, A., Jeon, H., Kee, T. R., Woo, J. A., & Kang, D. E. (2023). Disruption of Mitophagy Flux through the PARL-PINK1 Pathway by CHCHD10 Mutations or CHCHD10 Depletion. *Cells*, 12(24), 2781. <https://doi.org/10.3390/cells12242781>

**Wetzel, L.**, Mohammeda, Z.B., Yan, Y., Wang, X., Woo, J.A., Zhao, X., Zhu, Z., Liu, T. CHCHD10-Mediated MAM activity mitigates A $\beta$  generation and enhances A $\beta$  clearance in Alzheimer's Disease. *Manuscript*. 2024.

Sekyonda, Z., Oshabaheebwa, S., Zhang, B., **Wetzel, L.**, Wang, F., Fadaei, P., Du, Y., Man, Y., Chang, A., An, R., Liu, T., Baskaran, HA., Little, J., Gurkan, UA. (2025). New sickle cell disease therapies modulate red blood cell deformability via Ca<sup>2+</sup>, ATP, and 2,3-DPG in normoxia and hypoxia. *Blood Red Cells & Iron*, Volume 1, Issue 2, 2025, 100015. <https://doi.org/10.1016/j.brci.2025.100015>

Cheng, F., Qiu, Y., Hou, Y., **Wetzel, L.**, Caldwell, J., Zhu, X., Liu, T., Piper, A. Systematic characterization of cell type-specific master metabolic regulators in Alzheimer's disease. *Manuscript*. 2025.

Thomas, T., Yang, CY., **Wetzel, L.**, Zhang, K., Bugybayeva, D., Ferguson, J., Patel, N., Dash, A., Gowda, T., Zhu, X., Cheng, F., Liu, T. CHCHD10 alleviates Alzheimer's disease pathogenesis by regulating epigenetic landscape in directly reprogrammed neurons. *Submitted. Advanced Science*, 2025.

Hou, Y., **Wetzel, L.**, Chen, X., Yang, X., Bugybayeva, D., Thomas, T., Song, W., Liu, Z., Li, Y., Tan, Z., Hu, M., Li, YE., Zhu, X., Pillai, JA., Pieper, AA., Liu, T., Cummings J., Cheng, F. Leveraging cell type-specific genomic/epigenomic analyses and experimental validation identifies drug targets in neurodegenerative disease. **Submitted**. *Nature Communications*, 2025.

Hou, Y., **Wetzel, L.**, Chen, X., Yang, X., Bugybayeva, D., Thomas, T., Song, W., Liu, Z., Li, Y., Tan, Z., Hu, M., Li, YE., Zhu, X., Pillai, JA., Pieper, AA., Liu, T., Cummings J. Combining human genetics and cell type-specific x-QTLs identifies therapeutic targets in neurodegenerative disease. **Submitted**. *Nature Aging*, 2025.

Liu, T., **Wetzel, L.**, Roy, D., Zhang, K., Zhu, Z., Kumaraguru, P., Gorthi, V., Jeon, H., Woo, J.A., Kang, D.E. APP E590D mutation increases generation of A $\beta$  and A $\eta$  peptides and exacerbates tauopathy. *npj Dement.* 2, 21 (2026).  
<https://doi.org/10.1038/s44400-026-00069-9>

## Posters/Presentations

“CHCHD10 variants regulate mitophagy through PARL and impact TDP-43 accumulation.” 8th Annual Pathology Retreat. Case Western Reserve University School of Medicine, Cleveland, OH, October 2022.

“Protective role of mitochondrial CHCHD10 in tauopathy in AD.” 5th Annual International Research Conference on Neurodegenerative Diseases, Gainesville, FL, July 2025.

“MTCH2: a novel regulator of tau aggregation and neuroinflammation in AD.” 1st Annual Alzheimer’s Disease Translational Data Science Training Program Poster Symposium. Department of Population and Quantitative Health Sciences, Case Western Reserve University School of Medicine, Cleveland, OH, May 2026.

## Professional Laboratory Experience

### Case Western Reserve University – School of Medicine, Dept. of Pathology

Cleveland, OH

*Research Assistant II: June 2022 – July 2025*

- Supported research projects in the field of Alzheimer’s disease
- Performed wet bench procedures, cell culture, mouse colony management, and data management
- Conducted literature searches and order management
- Led undergraduate technicians through protocols and data analysis

## **University Hospitals - Cleveland Medical Center**

Cleveland, OH

**Histology Technician – Anatomical Pathology Slide Categorization:** *March 2022 – Sept. 2022*

- Independent assessment and filing of surgical and anatomical pathology slides
- Records and non-records management of PHI under HIPAA guidelines
- Coordinating and troubleshooting problems in file management systems
- Working with a dynamic team to provide efficient and timely return of non-UH samples

## **ALS Environmental**

Blue Ash, OH

**Laboratory Technician – Extraction (Soils/Waters):** *July 2021 – January 2022*

- Trained four (4) individuals in organic extraction methods
- Conducted assignments in Diesel Range Organics Gas Chromatography analysis
- Extracted organic compounds from soil and water samples via methylene chloride/hexane solutions, microwave procedure, decanting, and aliquoting
- Planned daily tasks and managed rush deadline orders
- Coordinated with other technicians to provide meticulous notes on batch execution and errors
- Organized and maintained a safe, clean, and effective laboratory environment
- Prepared samples for Agilent Gas Chromatography systems and chromatogram charts

## **Gravity Diagnostics**

Covington, KY

**Sample Management Associate I:** *November 2020 – July 2021*

- Receive and process samples in a timely manner.
- Accurately input data from requisition forms into Orchard Copia.
- Ensure patient information is secure and HIPAA compliant.
- Communicate effectively with team and leadership for an ever-changing healthcare environment
- Ensure proper development of document management and insurance billing resolution.

## Teaching & Advising Experience

### UNLV School of Life Sciences

Las Vegas, NV

**Undergraduate Teaching Assistant:** *August 2017 – December 2017*

- Guided students on laboratory safety, media preparation, and aseptic methods.
- Prepared 7 differential and structural stains; 16 differential, biochemical and physiological tests.
- Interpreted 6 medical diagnostics/application tests.
- Aided students' understanding in 3 microbial genetics concepts and procedures.
- Created a lab report identifying an unknown bacterium.

## Leadership & Academic Service

- The Global Association for the Study of Neurodegenerative Diseases - Trainee member (*July 2025 - present*)
- Association for Pre-Health Professionals – **Sergeant at Arms:** *Fall 2016 – Spring 2019*
- UNLV Pre-Dental Society: *Spring 2018 – Spring 2019*
- Health Improves Together – UNLV School of Medicine Outreach Program
- Western Psychological Association
- Association for Psychological Science
- Psi Chi and Psychology Club

## Continuing Education Credits

### MediaLab, Inc.

ACLS P.A.C.E Provider #578

Florida Board of Clinical Laboratory Personnel Provider #50-10293

Total Credits: 26 (in order of completion)

- OSHA Hazard Communication and Chemical Hygiene (P.A.C.E Contact Hours: 1)
- OSHA Fire Safety (P.A.C.E Contact Hours: 1)
- OSHA Electrical Safety (P.A.C.E Contact Hours: 1)
- OSHA Bloodborne Pathogens (P.A.C.E Contact Hours: 1.5)
- Medicare Compliance for Clinical Laboratories 2020 (P.A.C.E Contact Hours: 2.5)
- Medicare Compliance for Clinical Laboratories 2021 (P.A.C.E Contact Hours: 2.5)

- HIPAA Privacy and Security Rules 2020 (P.A.C.E Contact Hours: 2)
- HIPAA Privacy and Security Rules 2021 (P.A.C.E Contact Hours: 2)
- Vitamin D (P.A.C.E Contact Hours: 1)
- Bioterrorism: The Laboratory's Role and Response (P.A.C.E Contact Hours: 1.5)
- Overview of Prion Diseases (P.A.C.E Contact Hours: 1)
- Drug-Resistant Superbugs, Multi-Drug Resistant Organisms: MRSA, VRE, Clostridium difficile, and CRE (P.A.C.E Contact Hours: 2.5)
- Tuberculosis Awareness for Health Care Workers (P.A.C.E Contact Hours: 1)
- Microbial Identification Using MALDI-TOF MS (P.A.C.E Contact Hours: 1)
- COVID-19: Basics and Biosafety Precautions (P.A.C.E Contact Hours: 1.5)
- Routine Venipuncture (P.A.C.E Contact Hours: 1.5)
- Fundamentals of Molecular Diagnostics (P.A.C.E Contact Hours: 1)
- Alzheimer's Biomarkers: Overview of Existing and Future Biomarkers (P.A.C.E Contact Hours: 1.5)